

Total No. of Questions : 6]
P5039

SEAT No. :

[Total No. of Pages : 2

T.E./Insem. - 537
T.E. (Computer Engineering)
OPERATING SYSTEM DESIGNS
(2012 Pattern) (Semester - I)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) *Neat diagrams must be drawn wherever necessary.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data, if necessary.*

Q1) a) Explain the terms BIOS, MBR, GRUB in detail. **[6]**

b) Explain following algorithms. **[4]**

- i) iget
- ii) getblk

OR

Q2) a) What is buffer cache? Explain different buffer retrieval scenarios. **[5]**

b) Explain different file access methods with the help of suitable diagram. **[5]**

Q3) a) Consider the following snapshot of system at time T₀. **[8]**

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	2	1	0	0
P1	2	0	0	0	2	7	5	0				
P2	0	0	3	4	6	6	5	6				
P3	2	3	5	4	4	3	5	6				
P4	0	3	3	2	0	6	5	2				

P.T.O.

- i) Calculate the need matrix.
 - ii) State whether system is safe? If yes, give safe sequence (Show calculations).
 - iii) For the system snapshot at time T_0 , if a request from P_2 arrives for $(0,1, 0,0)$. Can the request be safely granted?
- b) What is shell? Enlist different types of shell. [2]

OR

- Q4)** a) What is difference between process and thread? State the advantages and disadvantages of user level threads. [5]
- b) What is deadlock? Describe necessary and sufficient conditions to occur deadlock. [5]

- Q5)** a) A process reference pages in the following order : [6]
5, 6, 7, 8, 5, 6, 9, 5, 6, 7, 8, 9

Use FIFO and LRU and Optimal page replacement algorithms to find out number of page faults for this reference string using 3 page frames.

- b) Explain internal and external fragmentation with suitable example. [4]

OR

- Q6)** a) Describe how address translation takes place (logical address to physical address) with the help of a neat diagram in [5]
i) Paging
ii) Segmentation

- b) Explain the following memory allocation strategies with suitable example: [5]
i) Best fit
ii) First fit
iii) Next fit

